

The ICT revolution, world economic growth, and policy issues[☆]



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ABSTRACT

The ICT revolution fueled by the exponential progress of the semiconductor technology and the accelerated pace of globalization has become an important driver of economic growth across nations. In this rapidly changing landscape, the world economy is entering into a New Economic Order, in which developing Asia led by two fast-growing giant economies, China and India, will have much larger impacts on the world economy. This paper provides empirical evidence on these phenomena and highlights policy issues that deem important for a country to seize the ICT revolution for promoting economic growth.

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1. Introduction

Information and Communications Technology (ICT) has brought about revolutionary changes in the way people work, communicate, learn, spend time, and interact. ICT has also profoundly transformed business and government practices. The impacts of the ICT revolution are now palpable in all countries, and are projected to be economically and socially revolutionary in the coming years as technology penetrates and fosters fundamental change in all sectors and dimensions of life.

At the heart of the ICT revolution is the rapid progress of semiconductors. The birth of modern ICT was marked by the invention of the transistor, a semiconductor device that acts as an electrical switch and encodes information in binary form. This takes the values zero and one, corresponding to the “off” and “on” positions of a switch. The first transistor, made of the semiconductor germanium, was constructed at Bell Labs in 1947 and won the Nobel Prize in Physics in 1956 for the inventors – John Bardeen, Walter Brattain, and William Shockley.¹

The second major milestone in the progress of ICT was the invention of the integrated circuit by Jack Kilby of Texas Instruments in 1958 and Robert Noyce of Fairchild Semiconductor in 1959. An integrated circuit consists of multiple transistors that store and manipulate data in binary form. Integrated circuits were originally developed for data storage and retrieval and semiconductor storage devices became known as memory chips. Kilby was awarded the Nobel Prize in Physics in 2000 for the invention of the integrated circuit; regrettably, Noyce died in 1990.²

The progress of semiconductor technology has been the driving force of the ICT revolution. In 1965 Gordon Moore, Research Director at Fairchild Semiconductor and later a founder of Intel in 1968, made a prescient observation, later known

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¹ On Bardeen, Brattain, and Shockley. See: http://www.nobelprize.org/nobel_prizes/physics/laureates/1956/.

² On Kilby. See: http://www.nobelprize.org/nobel_prizes/physics/laureates/2000/.

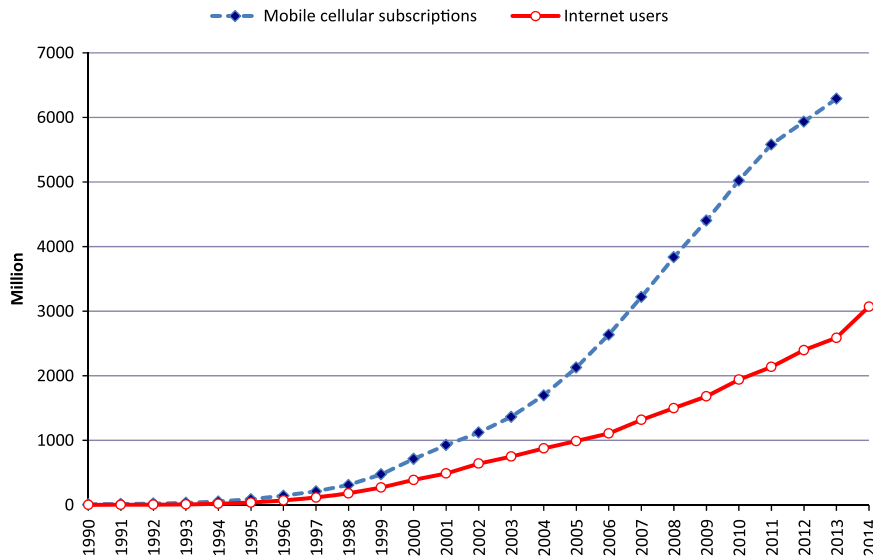


Fig. 1. World internet users and mobile phone subscribers, 1990–2014.

Data source: World Bank Development Indicators; see: <http://wdi.worldbank.org/tables>.

as Moore's Law: The number of transistors on a new chip would double within 18–24 months, which implied an exponential growth rate of 35–45 percent per year.³

In 1971 Intel created the Intel® 4004, the world's first central processing unit (CPU), which revolutionized the way computers are designed and applied. The CPU takes instructions from a stored program in read-only memory and works as the brain of a computer. The processing capability of a computer can be roughly assessed on the basis of the number of transistors on its CPU chip, which has followed Moore's Law since it was introduced.

For example, the number of transistors on Intel's CPU chip increased from 2250 in 1971 (the Intel® 4004 model) to 125 million in 2004 (Pentium 4 Prescott), 291 million in 2006 (the Core 2 Duo Conroe model), and 5560 million in 2014 (the 18-core Xeon Haswell-E5 model).⁴ This implies an annual growth rate of 41 percent over 1971–2014. Moore has forecast that the Moore Law will continue to hold for the next 10 years.⁵

Just as the innovation in semiconductors and personal computers has established the foundation for the progress of ICT, the emergence of the internet and mobile technology in the 1990s, together with the accelerated pace of globalization, has fueled the dramatic spread of applications of ICT across sectors and nations. In Fig. 1 we show that the world's internet users soared from three million in 1990 to more than three billion in 2014. The number of mobile phone subscribers increased from 11 million to more than six billion over the same period.

The spread of ICT in developing countries has been even more remarkable. The penetration of internet and mobile phone went from nil in the early 1990s to impressively high rates even in the poorest and most isolated nations.⁶ The spread of ICT into all corners of the world has had a profound effect on economic development, particularly in areas where communication, access to information, learning, research, and innovation play a key role in driving success. The investment in ICT, therefore, has had measureable effects on economic growth for all nations.

In Section 2 we present empirical evidence on the contribution of ICT to economic growth in the past two decades, from 1990 to 2012. In Section 3 we analyze the future impact of ICT on world economic growth by presenting intermediate-term projections for the world economy, regions of the world, and major developed and emerging economies. In Section 4 we consider the implications of ICT for economic policy.

³ Intel (2015): <http://www.intel.com/content/www/us/en/silicon-innovations/moores-law-technology.html?wapkw=moore%ca%b9s+law>.

⁴ Intel had to change its strategy of increasing the number of transistors on a single processor to raise the computing power of CPU in 2004 when it hit "a thermal wall" caused by energy consumption inefficiency. The company then followed the multiple core strategy, which stamps multiple processors on a single CPU chip to increase its number of transistors. This development of Intel microprocessors is discussed by Markoff (2004). See: <http://www.nytimes.com/2004/05/17/business/technology-intel-s-big-shift-after-hitting-technical-wall.html>.

⁵ Moore (2015): <http://venturebeat.com/2015/05/11/intels-gordon-moore-speculates-on-the-future-and-the-end-of-moores-law/>.

⁶ For example, in 2013 the penetration rate of mobile phones (per 100 inhabitants) was 9.7 in North Korea, 12.8 in Myanmar, 25.3 in South Sudan, and 27.3 in Ethiopia, which were comparable with the rates observed for the industrialized countries in early 1990s.

2. Contribution of ICT investment to economic growth

The rapid spread of mobile phones, the internet, and other applications of ICT has spurred sizable investments in ICT assets. This section estimates the contribution of ICT investment to growth of the world economy over the period from 1990 to 2012. For this purpose, we adopt the model originated by Jorgenson, Ho, and Stiroh (2008) to decompose the aggregate growth into the contribution of inputs of capital and labor services and total factor productivity (TFP), defined as output per unit of all inputs.

In the production function (1), output (Y) is produced by an input bundle X of capital services rendered by ICT capital (K_{ICT}) and non-ICT capital (K_{NICT}), and labor services comprised of hours worked (H) and labor quality (L_Q). Total input (X) is augmented by total factor productivity (A) or output per unit of inputs of both capital and labor services:

$$Y = AX(K_{ICT}, K_{NICT}, H, L_Q) \quad (1)$$

Under the assumption of perfect competitive factor markets where the marginal product of each input equals its price and constant returns to scale, Eq. (1) can be transformed into the growth accounting Eq. (2)

$$\begin{aligned} \Delta \ln Y &= \bar{\nu}_K \Delta \ln K + \bar{\nu}_L \Delta \ln L + \Delta \ln A \\ &= \bar{\nu}_{K_{ICT}} \Delta \ln K_{ICT} + \bar{\nu}_{K_{NICT}} \Delta \ln K_{NICT} + \bar{\nu}_L \Delta \ln H + \bar{\nu}_L \Delta \ln L_Q + \Delta \ln A \end{aligned} \quad (2)$$

where $\bar{\nu}$ denotes the two-period average shares of total factor income, and the change in the logarithm $\Delta \ln$ of a variable denotes its real growth rate. The assumption of constant returns to scale of the aggregate input function implies that $\bar{\nu}_K + \bar{\nu}_L = 1$ ($\bar{\nu}_K = \bar{\nu}_{K_{ICT}} + \bar{\nu}_{K_{NICT}}$). The growth accounting Eq. (2) implies that GDP growth can be decomposed into three main sources:

- Contribution of capital inputs, which consists of the contributions of ICT and non-ICT capital: $\bar{\nu}_K \Delta \ln K = \bar{\nu}_{K_{ICT}} \Delta \ln K_{ICT} + \bar{\nu}_{K_{NICT}} \Delta \ln K_{NICT}$,
- Contribution of labor input, which comprises the contributions of hours worked and labor quality: $\bar{\nu}_L \Delta \ln L = \bar{\nu}_L \Delta \ln H + \bar{\nu}_L \Delta \ln L_Q$, and
- Contribution of TFP growth, $\Delta \ln A$

A key feature of the growth accounting framework for the sources of economic growth is that capital input is defined in terms of capital services rather than capital stock.⁷ The prices of capital inputs used in aggregating different types of capital are rental values rather than asset values. This approach to measuring capital was incorporated into official statistical systems in 2009. The change from capital stock to capital services has been crucial to estimating the impact of the shift in the composition of capital toward information technology equipment and software.

Our measure of labor input incorporates investment in human capital in the form of education and experience. In addition we include differences in hours worked by workers with different demographic characteristics, such as age, sex, and education. Labor quality is the ratio of labor input to hours worked, so that labor input is the product of hours worked and labor quality. Labor quality is crucial for capturing the impact of investment in human capital on economic growth.

Productivity (TFP) is essential for dividing the sources of economic growth between the two broad categories of replication and innovation. *Replication* takes place by adding identical production units with no change in technology. Labor input grows through the addition of new members of the labor force with the same education and experience and capital input expands by providing new production units with the same collection of plant and equipment. Output expands in proportion to the capital and labor inputs with no change in productivity. By contrast, *innovation* involves new products and new processes, so that productivity can increase. The distinction between replication and innovation will be central to our analysis of the sources of economic growth.

In analyzing the growth of the world economy we use the Total Economy Database (TED) that we have developed with The Conference Board in New York. The database is updated annually by The Conference Board, for which the sources and methods are presented in Vries and Erumban (2015). We have published a series of international comparisons of outputs, inputs of capital and labor services, and productivity – the sources economic growth – based on TED.⁸

Data on capital and labor inputs are essential in analyzing recent data on economic growth, preparing intermediate-term projections of future growth, and choosing among alternative economic policies. The framework for our growth projections is provided by the neo-classical theory of economic growth.⁹ Productivity growth provides a link to technical change. Growth in labor inputs is based on demographic projections from the United Nations and by national statistical agencies like the U.S. Bureau of the Census.¹⁰ Growth in capital inputs is determined endogenously by saving and investment behavior.

⁷ For a given asset type, capital services (productive generated by the asset) instead of capital stock (accumulative investment) is the appropriate measure of capital input for production and productivity analysis. See Schreyer (2001, 2009) for elaboration on this methodology.

⁸ For example, see Jorgenson and Vu (2013).

⁹ See Jorgenson, Ho, and Stiroh (2008). http://scholar.harvard.edu/files/jorgenson/files/retrosprctivelookusprodgrowthresurg_journaleconperspectives.pdf.

¹⁰ <http://esa.un.org/wpp/> <http://www.census.gov/population/international/data/idb/informationGateway.php>.

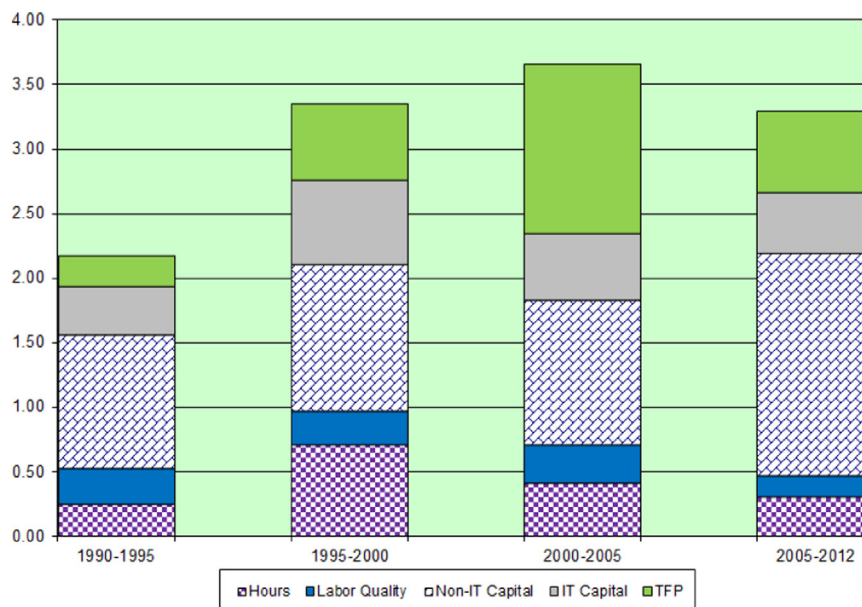


Fig. 2. Sources of world economic growth, 1995–2012 (annual percentage growth rates).
Source: Authors' calculation.

The Total Economy Database incorporates the recent advances in growth accounting summarized in Paul Schreyer's *OECD Productivity Manual* of 2001 and his *OECD Capital Manual* of 2009.¹¹ International standards were established in Chapter 20 of the United Nations' 2008 *System of National Accounts* (SNA), published in 2009.¹² This is the most recent revision of the United Nations' SNA and was developed in collaboration with four other international organizations – the OECD, the IMF, the World Bank, and Eurostat.

In Fig. 2 we divide the contribution of capital services to world economic growth between ICT and Non-ICT assets. We allocate the contribution of labor services between hours worked and labor quality, defined as labor input per hour worked. Finally, we present data on total factor productivity, defined as output per unit of both capital and labor inputs.

In order to prepare for our projections of future world economic growth we analyze the sources of economic growth for the G7, the G20, and the world economy.¹³ We consider 14 major economies – the G7 economies, including the U.S., and seven emerging economies of the G20, including India and China. All comparisons in this paper are based on GDP measured in purchasing power parity (PPP) dollars, provided by the World Bank Comparison Program (ICP) 2011. The aggregation for group and world is based on the country GDP weight measured in PPP.

We have sub-divided the period 1990–2012 into 1995, 2000, and 2005 in order to capture major changes in the growth rate of the world economy. The period 1995–2000 witnessed a major acceleration in world economic growth. World growth accelerated further during 2000–2005. The period after 2005 was dominated by the economic and financial crisis of 2007–2009 in the advanced economies that comprise a major proportion of the world economy. World economic growth continued at a rapid pace, but slowed relative to 2000–2005.

In measuring capital input, we distinguish between information technology equipment and software, which we call ICT capital, and non-ICT capital. Second, we divide the contribution of labor input between hours worked and labor quality. Hours worked reflects the mobilization of labor by increased labor force participation and higher rates of utilization of labor. Labor quality captures the contribution of human capital, accumulated through education and experience. For the world economy the growth of labor quality exceeded growth of hours worked in 1990–1995, but hours worked have grown more rapidly than labor quality since 1995.

Total factor productivity growth for the advanced economies of the G7 was negative for 2005–2012, except for Germany, Japan and the U.S. This poses a paradox: How could innovation, reflecting changes in technology, be negative? The world-wide financial and economic crisis of 2007–2009 opened a wide gap between actual output and potential output for the G7 economies, determined by supplies of capital and labor inputs and productivity.¹⁴ The actual level of output reflects the

¹¹ <http://www.oecd.org/std/productivity-stats/2352458.pdf>; <http://www.oecd.org/std/productivity-stats/43734711.pdf>.

¹² <http://unstats.un.org/unsd/nationalaccount/sna2008.asp>.

¹³ The seven largest industrialized economies (G7) are the U.S., Japan, Germany, France, the U.K., Italy, and Canada. The G20 group contains 20 largest economies, which includes the G7 economies, the non-G7 European Union, the seven largest emerging economies (China, India, Russia, Brazil, Mexico, South Korea, and Indonesia, which will hereafter be referred to as E7 economies), and five other large economies (Turkey, Australia, Argentina, South Africa, and Saudi Arabia).

¹⁴ OECD (2011) put the output gap of the OECD area at 0.7% (of potential GDP) as an average for 1999–2008, –4.4% for 2009, –3.2% for 2010, –3.1% for 2011, and –3.4% for 2012. For the U.S., these respective figures are 1.4%, –4.9%, –3.7%, –3.8%, and –4.0% (Annex Table 10, p. 232). Details on the OECD projection methods and underlying statistical concepts can be found in OECD Economic Outlook Sources and Methods. <http://www.oecd.org/eco/sources-and-methods.htm>.

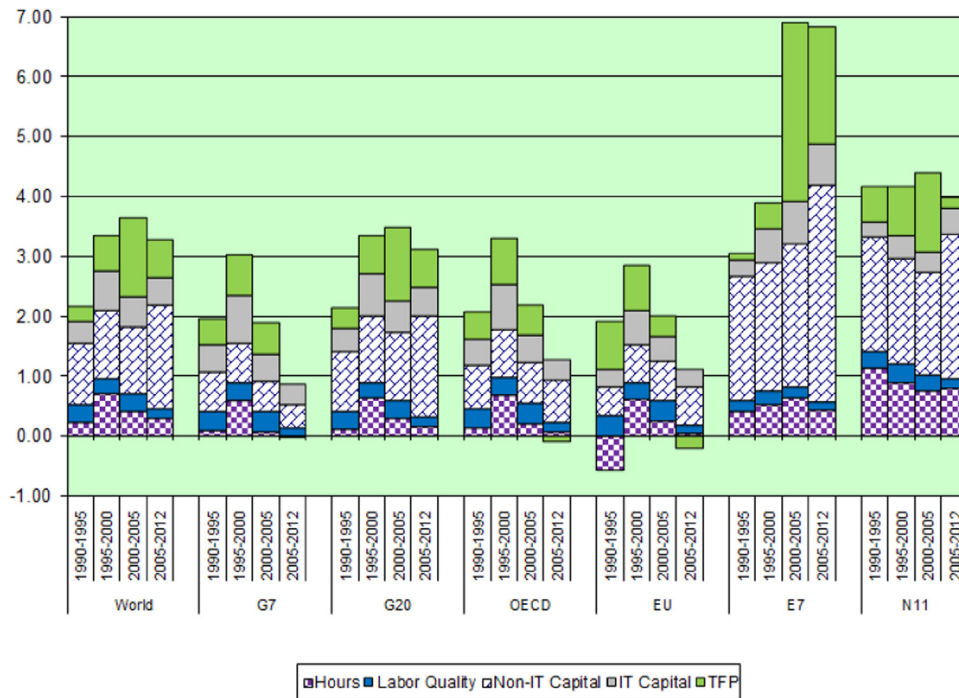


Fig. 3. Sources of world economic growth, 1990–2012: major groups (annual percentage growth rates). *Notes:* The EU group includes 25 economies: Austria, Belgium, Bulgaria, Czech, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Netherlands, Poland, Portugal, Romania, Slovak, Slovenia, Spain, Sweden, and the U.K.; The E7 (Emerging 7) group consists of seven largest emerging economies: China, India, Russia, Brazil, Mexico, South Korea, and Indonesia; The N11 (Next 11) comprises 11 emerging large economies: Bangladesh, Egypt, Indonesia, Iran, Mexico, Nigeria, Pakistan, Philippines, South Korea, Turkey, and Vietnam.

Source: authors' calculation.

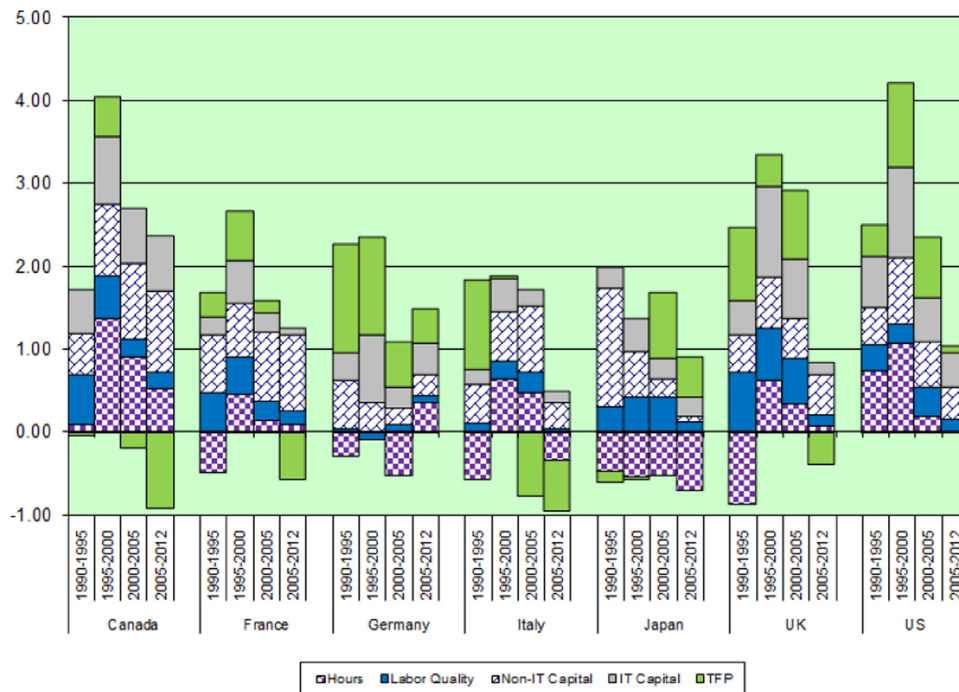


Fig. 4. Sources of economic growth, 1990–2012: G7 economies (annual percentage growth rates).

Source: Authors' calculation.

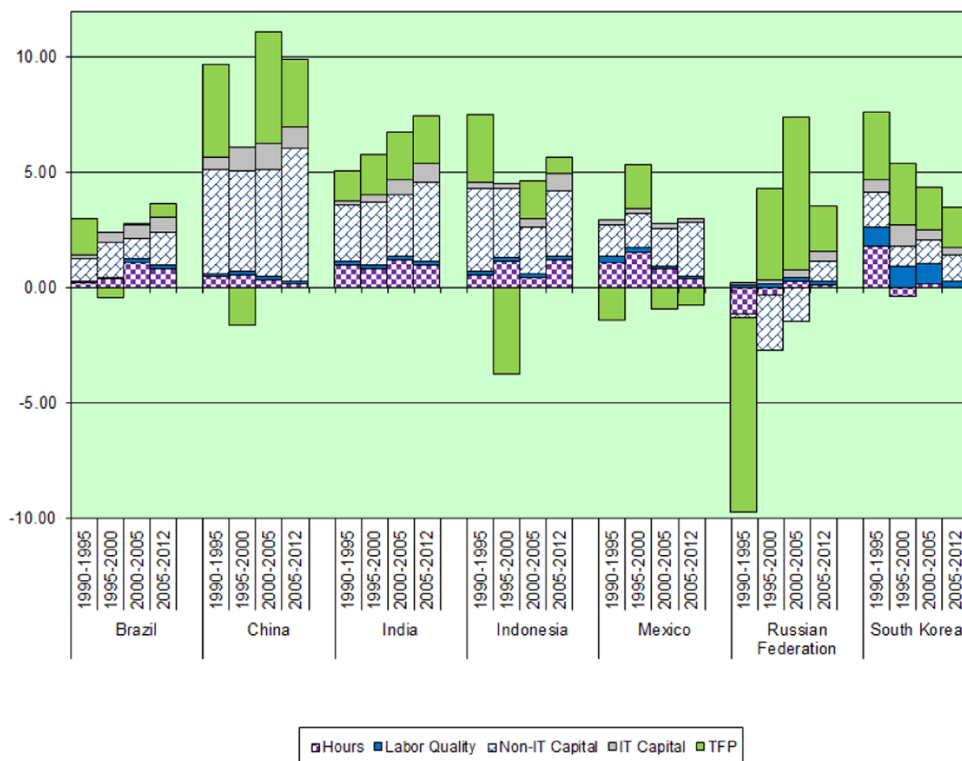


Fig. 5. Sources of economic growth, 1990–2012: E7 Economies (annual percentage growth rates).

Source: Authors' calculation.

deficiency of aggregate demand that accompanied the crisis. Negative total factor productivity growth is the difference between the positive effects of advances in technology and the negative effects of the widening output gap.

During the period 1990–2012 U.S. economic growth has been strong, relative to other advanced economies (Fig. 3). This is due to capital deepening, increases in capital input per hour worked, especially during the information technology boom of 1995–2000. Growth in total factor productivity revived during the boom and continued at a brisk pace in the “jobless recovery” of 2000–2005. The growth of U.S. GDP has been more rapid than that of any other economy in the G7 with only Canada close behind. However, U.S. economic growth collapsed during the period 2005–2012, largely due to the emergence of a gap between potential and actual GDP during the Great Recession.

Like the U.S., the advanced economies of the G7 were strongly impacted by the Great Recession of 2007–2009. In Fig. 4 we analyze the sources of economic growth of the G7 during 1990–2012. Among the members of the G7 Germany had the strongest economic performance during the Great Recession with positive growth in total factor productivity and hours worked. Japan's productivity growth revived during the period 2000–2005 and this continued during 2005–2012. The revival of productivity growth was largely offset by a declining labor force throughout the period 1990–2012. Italy had the weakest performance during the Great Recession and was unable to maintain positive economic growth.

China sustained double-digit economic growth throughout the decade 2000–2012 with only a slight deceleration during the Great Recession, as shown in Fig. 5. India's growth rate rose steadily during the two decades 1990–2012. For both China and India capital input contributed more to economic growth than total factor productivity growth during these decades. Russia's economic growth during this period was dominated by the collapse of the former Soviet Union and the gradual recovery of 1995–2005. The decline in growth of Non-ICT capital input from 1990 to 2005 is an unusual feature of Russia economic growth that was finally reversed after 2005.

3. Projecting the growth of the world economy

To set the stage for our intermediate-term projections of future world economic growth, we summarize three important findings¹⁵: The first is that *world economic growth will accelerate during the decade 2012–2022*, relative to the past two

¹⁵ These findings are based on the projections derived from the base scenario. Our projections, to a large extent, are based on the growth pattern observed for each individual country in the past 10 years. As a result, our projections for China may be upward-biased because of two reasons: (i) China's growth in the recent years, especially since the global financial crisis has been heavily driven by government fiscal stimulation and (ii) China's data and

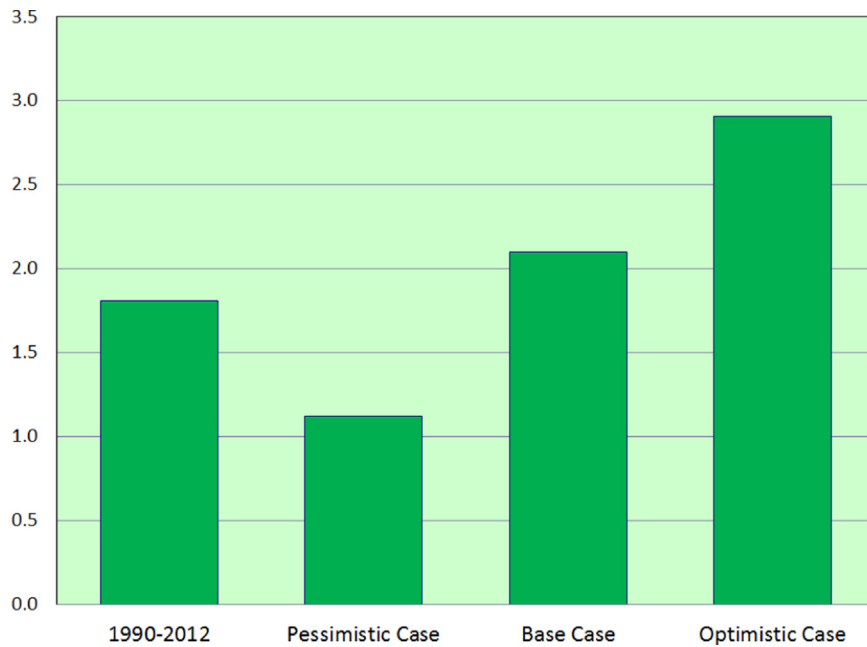


Fig. 6. Range of world labor productivity growth projections, 2012–2022 (annual percentage growth rates).
Source: Authors' calculation.

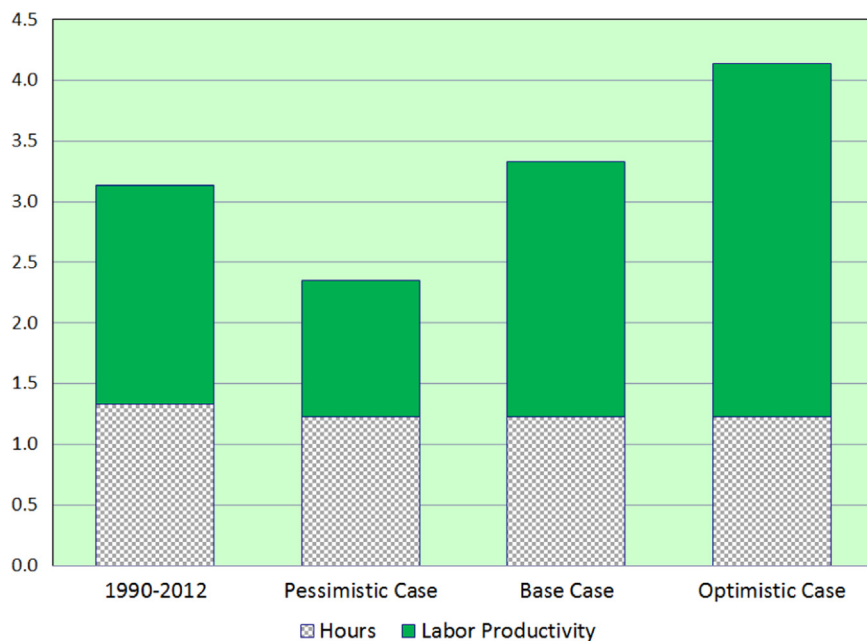


Fig. 7. Range of world potential output growth projections, 2012–2022 (annual percentage growth rate).
Source: Authors' calculation.

decades 1990–2012. While the recovery of the U.S. has recently slowed, Europe is emerging from recession and the financial and fiscal crisis is abating. However, economic growth in the G7 economies will encounter the formidable headwind of a decline in the labor force in Japan and much of Europe.

(footnote continued)

estimation on growth are subject to further revision to enhance their reliability. In his latest paper with the Conference Board, Wu (2014) shows that China's actual GDP growth during 1977–2012 was lower than its official rate by 1.8% points and this gap widened to 2% points for the recent period 2007–2012 (Table 9, p. 45).

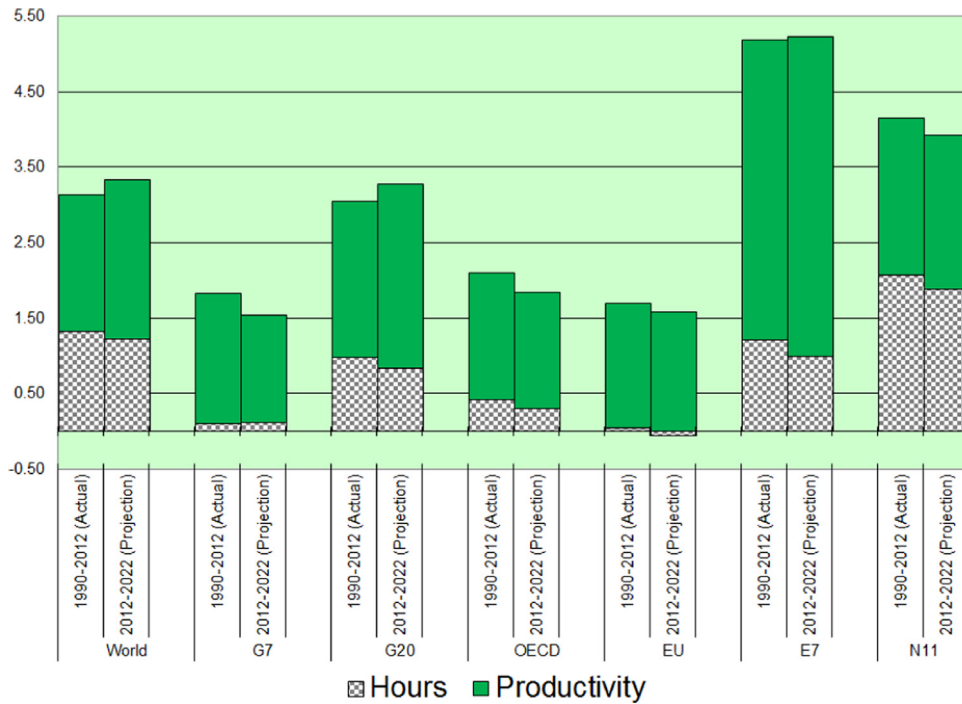


Fig. 8. Range of potential output growth projections, 2012–2022: Major groups (annual percentage growth rates). Source: Authors' calculation.

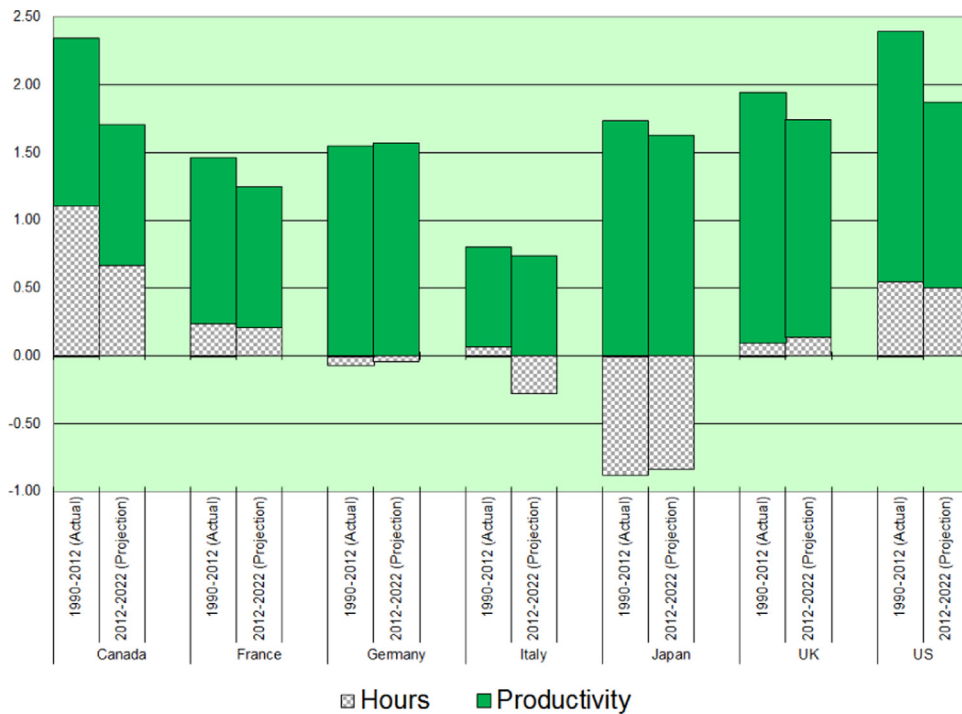


Fig. 9. Range of potential output growth projections, 2012–2022: G7 economies (annual percentage growth rates). Source: Authors' calculation.

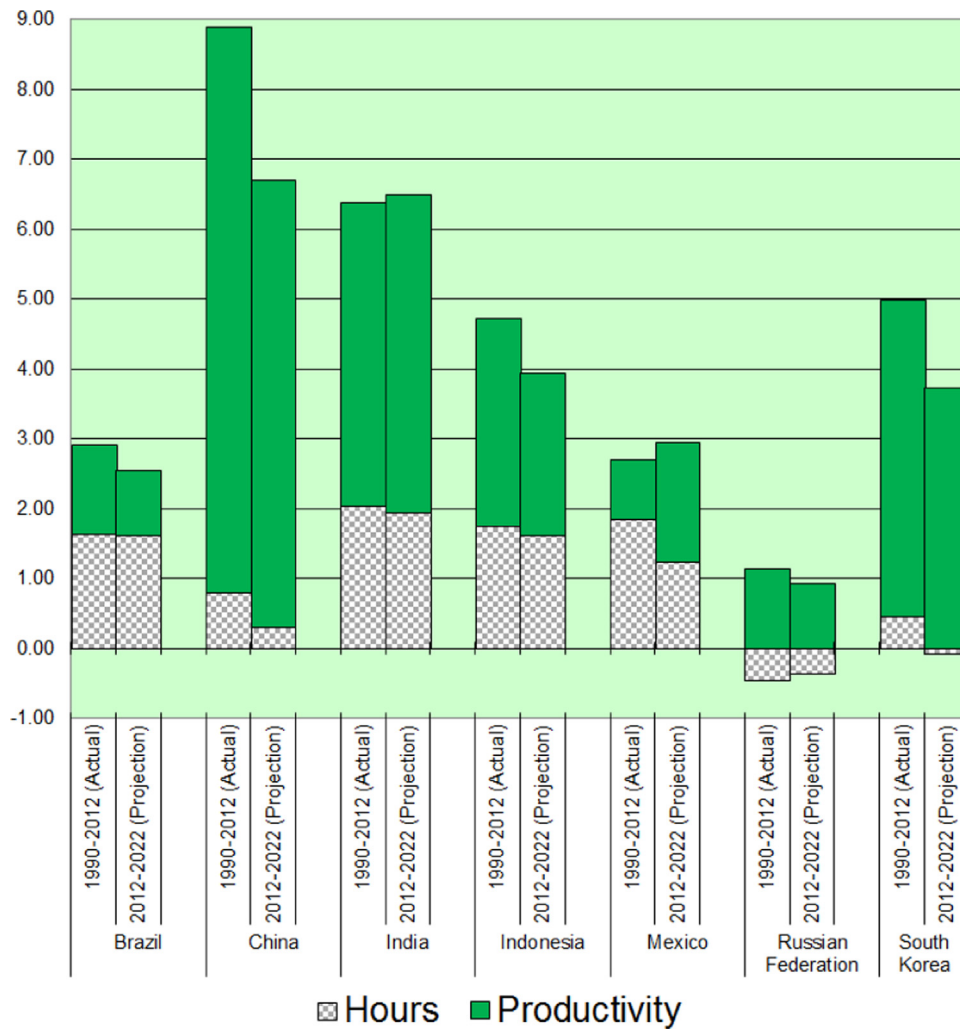


Fig. 10. Range of potential output growth projections, 2012–2022: E7 economies (annual percentage growth rates).
Source: Authors' calculation.

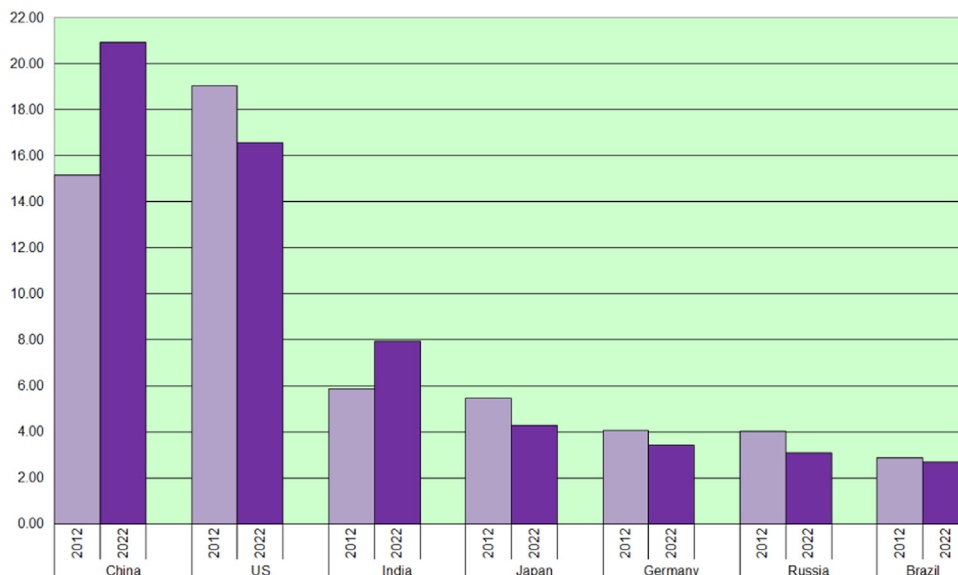


Fig. 11. New economic order, 2022 (percentage shares of world GDP).
Source: Authors' calculation.

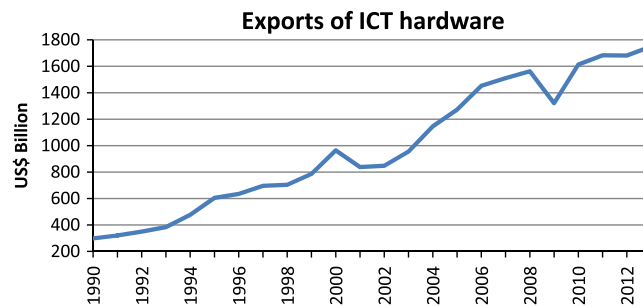


Fig. 12. Trade in ICT hardware.

Data source: WTO. See: [http://stat.wto.org/Home/WSDBHome.aspx?Language=.](http://stat.wto.org/Home/WSDBHome.aspx?Language=)

During 2012–2022 Chinese growth will slow relative to the rapid pace of the past two decades. However, growth in India will accelerate modestly. Both economies will continue to grow much more rapidly than the world economy. As these economies increase in relative importance, the growth rate of the world economy will rise. Overall, the growth rates of the G20 economies will be almost equal to that of the world economy, so that the G20 will account for about eighty percent of world GDP at the beginning and end of the decade of 2012–2022.

The second major trend in the world economy over the next decade is that *China will overtake the U.S. and India will overtake Japan*. One of the most heralded economic developments of 2010 was that China overtook Japan as the world's second largest economy in terms of exchange rates. According to the World Bank' 2005 International Comparison Program, China had overtaken Japan more than 5 years earlier in terms of purchasing power parities.¹⁶ In 2012 India overtook Japan in terms of purchasing power parities and will continue to grow much more rapidly.

China is growing much more rapidly than the U.S. and will continue to do so. When will China become No. 1? The World Bank's 2011 International Comparison Program calls for parity between the two economies in 2014 in terms of purchasing power.¹⁷ This has led to considerable debate among participants in ICP2011, including the Chinese participants. However, the U.S. and China made up more than a third of the world economy in 2012 and this proportion will increase by 2022, consolidating the rise of China and the positions of China and the U.S. as the world's leading economies.

The final intermediate-term developments are that Germany will continue to lead Russia and Russia, in turn, will lead Brazil, resulting in a *New Economic Order in 2022*: China, the U.S., India, Japan, Germany, Russia, and Brazil. The emergence of Brazil, Russia, India, and China confirms the prediction in 2001 by Jim O'Neill, then a Goldman–Sachs economist, about that future power of these four economies known as BRIC.¹⁸ In this paper, we consider a group of seven largest emerging economies (hereafter to be referred to as E7), which, in addition to the four BRIC countries, includes Mexico, South Korea, and Indonesia.¹⁹

In our projections, we consider three different scenarios for the growth of the world economy – Base Case, Pessimistic, and Optimistic. The contributions of hours worked and labor quality to potential economic growth over the next decade can be projected with little uncertainty, since most of the people entering the labor force are already in the population. Accordingly, we use the same projections in our alternative scenarios. Projections of technology are far more challenging. Our Base Case incorporates the average rate of total productivity growth for 1990–2012. Our Pessimistic Case is based on average productivity growth for the two worst sub-periods, while our Optimistic Case is based on an average for the two best sub-periods. For emerging economies like China and India we also extrapolate recent trends in the capital-output ratio.

We limit our projections to potential economic growth, leaving projections of actual growth to economic forecasters like the International Monetary Fund (IMF) and the World Bank.²⁰ We have incorporated differences in trends in demography and technology. For example, the growth of the Japanese population and labor force will continue to be negative and France, Germany, Italy, Russia and South Korea will also experience negative labor force growth. The growth of the Chinese labor force will decline substantially and India's labor force will grow much more rapidly than China's.

We have represented the uncertainty in our projections of the world economy for 2012–2022 in Figs. 6 and 7. The first bar in this Fig. 6 provides historical data. The average growth rate of labor productivity for 1990–2012 was 1.8 percent per year. The remaining bars give Pessimistic, Base Case, and Optimistic projections of labor productivity for the decade 2012–2022.

¹⁶ World Bank (2008). *2005 International Comparison Program*; see: http://siteresources.worldbank.org/ICPINT/Resources/ICP_final-results.pdf.

¹⁷ World Bank (2014). *2011 International Comparison Program*; see: http://siteresources.worldbank.org/ICPEXT/Resources/ICP_2011.html.

¹⁸ See Jim O'Neill (2001).

¹⁹ Jim O'Neill (2011) extends the BRIC into BRICS to include South Africa, a member of the G20 but not one of the largest seven emerging economies (in terms of GDP measured in purchasing power parity dollars, South Africa is twice smaller than Indonesia – the smallest economy in the E7 group).

²⁰ The IMF (for example, see IMF (2015)) projects actual growth in a given economy, taking into account expected stimulation/reform efforts of its government and the growth prospects of the major economies. Our projections focus on potential growth, which is solely determined by two supply-side factors within each country – productivity and employment.

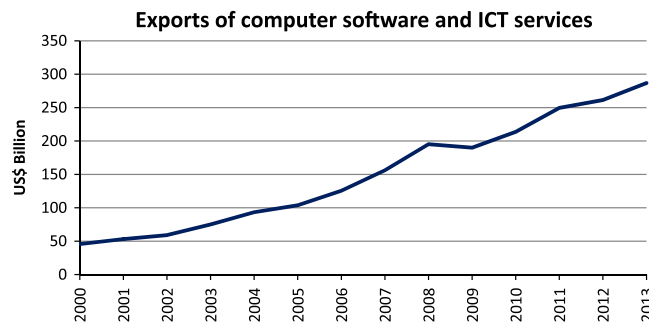


Fig. 13. Trade in ICT services.

Data source: WTO. See: [http://stat.wto.org/Home/WSDBHome.aspx?Language=.](http://stat.wto.org/Home/WSDBHome.aspx?Language=)

Table 1

Distribution of Apple's suppliers by country, 2014.

	Country	Suppliers	Share in total (%)
1	USA	60	7.6
	Asia	662	84.3
2	China	349	44.5
3	Japan	139	17.7
4	Taiwan	42	5.4
5	South Korea	32	4.1
6	Philippines	24	3.1
7	Malaysia	21	2.7
8	Thailand	21	2.7
9	Singapore	17	2.2
10	Vietnam	11	1.4
11	Indonesia	6	0.8
	Rest of world	63	8.0
12	Germany	13	1.7
13	Mexico	7	0.9
14	Israel	6	0.8
15	Czech	5	0.6
16	France	5	0.6
17	Belgium	3	0.4
18	Ireland	3	0.4
19	Italy	3	0.4
20	UK	3	0.4
21	Austria	2	0.3
22	Brazil	2	0.3
23	Costa Rica	2	0.3
24	Netherlands	2	0.3
25	Canada	1	0.1
26	Hungary	1	0.1
27	Malta	1	0.1
28	Morocco	1	0.1
29	Portugal	1	0.1
30	Puerto Rico	1	0.1
31	Spain	1	0.1
	Total	785	100.0

Source: Supplier list 2014, Apple's Supplier Responsibility Program. www.apple.com/supplierresponsibility.

More rapid productivity growth is associated with higher rates of growth of capital deepening, defined as the growth rate of capital per hour worked, as well as more rapid growth in total factor productivity. We project the rate of GDP growth as the sum of the growth rate of hours worked from our demographic projections and the growth rate of labor productivity. The first bar in Fig. 6 gives historical data for 1990–2012. The remaining three bars give Pessimistic, Base Case, and Optimistic projections.

Our Base Case projection of world economic growth for the period 2012–2022 is 3.4 percent per year. This corresponds to labor productivity growth of 2.2 percent per year and growth of hours worked of 1.2 percent. The Base Case reflects an acceleration of world economic growth, combining a slowdown in the growth of hours worked and a substantial acceleration in labor productivity growth. The Pessimistic projection of world growth of 2.4 percent per year incorporates a

Table 2

The seven dimensions of a typical ICT policy framework.

Dimension	Focused areas for promotion, upgrading, and progress monitoring
(i) ICT connectivity and access	• Broadband connectivity; • Mobile phone coverage;
(ii) ICT usage	• Cost and quality competition among ICT products and services providers • Use by individual; use by business; and use by government; • Incentives to promote ICT usage by the poor, rural communities, and SMEs
(iii) ICT legal and regulatory framework	• Telecommunications regulation; • Spectrum frequency allocation • E-commerce laws (digital signatures, intellectual property laws, e-payment); • Cybersecurity laws and regulations; • ICT trade tariff and regulations; • Access to data and cross-border data transfer; • Investment regulations
(iv) ICT production and trade	• ICT-enabled services and ICT content provision; • SMEs in the ICT sector (financing, investment, and capacity building); • E-market places; • Innovation and R&D in the ICT sector; • Special industrial parks/zones/villages for ICT sector development (such as software, high-tech, call centers)
(v) ICT skills and human resources	• ICT skills in primary and secondary schools; • ICT graduates and programs at universities and vocational schools; • Training projects to enhance the ICT workforce; • Incentives for private sector companies to organize/support ICT capacity building.
(vi) Cybersecurity	• Minimization of vulnerability to cyber-attacks; • National Cyber Security Exercise programs; • Emergency Response to Cyber-Attack plans
(vii) New ICT applications	• Smart city development; • Big data analysis; • Internet of things

Sources: Adapted from UNCTAD (2014); see: <http://unctad.org/en/pages/PublicationWebflyer.aspx?publicationid=880>; World Bank (2015): <http://www.worldbank.org/en/topic/ict>; Microsoft (2015): <http://pctechmag.com/2015/04/microsoft-partners-with-kenyas-icta-to-strengthen-ict-and-innovation/>.

continuation of the marked slowdown in growth of total factor productivity after 2005. The Optimistic projection for world economic growth of 4.1 percent would require full recovery from the Great Recession.

We present projections of economic growth for the world economy and major groupings of economies, including the G7, the G20, and the E7. We provide historical data for the period 1990–2012 and our Base Case projections for 2012–2022. The future growth of the world economy will accelerate, relative to the historical period. In Fig. 8 the advanced economies of the G7, the OECD, and the EU will grow more slowly, while the growth rate of the extended BRICs is almost the same. The modest acceleration in world economic growth is due to the rising importance of the more rapidly growing regions. Space limitations prevent us from indicating the substantial uncertainty associated with these projections.

Our projections for the individual economies in the G7 are given in Fig. 9. The U.S. will continue to outperform the other members of the G7 due to expansion of the labor force and growth of labor productivity. Labor productivity growth in Germany, Japan, and the U.K. will be stronger than in the U.S., but the rate of decline in the Japanese labor force will reach almost one percent per year. The German labor force will also decline, but less rapidly, while the U.K. labor force will increase only slightly. Italy will continue to lag in economic growth among the G7 economies, but Italian growth will be positive for 2012–2022. Canada's growth will be similar to the U.S., but growth of the Canadian labor force will drop considerably.

As shown in Fig. 10, the outlook for China and other emerging economies continues to be very sanguine, but quite different. The Chinese growth rate will decline relative to the blistering pace of the past decades, still exceeding 6.5 percent per year. Growth in India and Mexico will accelerate, while Brazil, Indonesia, Russia, and South Korea will grow more slowly. How can the world economy grow more rapidly? This is due to the rising importance of the more rapidly growing economies, especially China, but also India.

Fig. 11 summarizes the New Economic Order that will emerge by 2022. China will displace the U.S. as the world's leading economy, but China and the U.S., taken together, will increase in relative importance in the world economy. India's growth rate will rise to 6.5 percent, so that India will also increase in relative importance but Indian growth will be insufficient to avoid a considerable fiscal consolidation. Russian economic growth will continue to languish and Russia will fail to overtake Germany. Brazil will grow more rapidly than Russia, but Brazilian GDP will remain below the GDP of Russia.

It is important to emphasize that there are substantial uncertainties for each of the world's leading economies. For example, China's continuing accumulation of foreign exchange to preserve the dollar exchange rate of the Chinese currency is reaching a plateau. The real exchange rate of the yuan is rising due to a modest revaluation and a very substantial increase in production costs in China. All of this is helping to moderate the imbalances that have attracted international attention. China is moving toward slower, but more sustainable growth.

Additional evidence on the contribution of ICT-producing industries to growth: Due to data limitations, it is difficult to estimate the contribution of the ICT-producing sector to growth on a global basis. This section, therefore, only aims to shed some light into the contribution of the sector by examining two issues. The first is the increasingly global fragmentation of the ICT hardware production process. The second is the distribution of value captured by the countries participating in this global supply chains.

The global ICT-producing sector has experienced not only remarkable growth but also substantial redeployment as companies seize the globalization trend to enhance their cost-saving, efficiency, and market responsiveness. In particular, the labor-intensive activities in this production process have been shifted to or allocated in low labor cost countries such as China and Southeast Asia (for hardware) and India (for software and ICT services). As a result, the international trade of ICT products, of which a large portion consists of intermediate inputs, has soared at rapid rates over the recent decades.

World exports of computer and telecommunication hardware products increased by nearly six times over 1990–2013, from US\$298 billion in 1990 to 1750 billion in 2013 (Fig. 12), while exports of ICT services (data available only from 2000) rose by more than six times over 2000–2013, from US\$46 billion in 2000 to 287 billion in 2013 (Fig. 13). It is worth noticing that China, which has become a global hub of ICT hardware manufacturing, increasing its export value of ICT hardware by nearly 200 times over 1990–2013 from US\$3 billion in 1990 to US\$594 billion in 2013. At the same time, India has become the world's largest exporter of computer software and ICT services, increasing its export value by more than 12 times over 2000–2013, from US\$4 billion in 2000 to 50 billion in 2013²¹.

The worldwide distribution of Apple's suppliers in 2014 provides an illuminating example of the global fragmentation of the ICT hardware production process. Its suppliers spread over 31 countries, of which 85 percent are located in Asia (Table 1). It should be noted that while most labor-intensive activities in the ICT hardware production process have shifted to China and other South East Asian, the distribution of the value captured from this value chain is heavily skewed towards the countries with higher intellectual contributions.

Kraemer, Linden, & Dedrick (2011) provide an illuminating example through the case of Apple's iPhone4. According to their estimates, Apple captured 58.5 percent of the sales of the iPhone4, while the cost of material inputs accounted for approximately 22 percent. South Korea claimed 4.7 percent, while Japanese, European, and other non-Apple US players captured 0.5–2 percent each. China, which assembled the final product, earned about 2 percent from labor compensation.²²

4. Policy discussion

The ICT revolution has had profound impacts on economic development in all dimensions. As presented in the previous sections, the contributions of ICT to economic growth are empirically measurable. This section highlights policy issues that can help a nation reap more benefits from the ICT revolution. The main focus of this section is the policies for promoting ICT usage.

Policies related to ICT must consider not only its management and regulation, but also the broader spheres of impact on economy, society, and governance. With the emergence of ICT in the past recent decades, governments have sought to produce coherent, effective policies ahead of technology development trends. Such trends typically outpace the ability of policies to anticipate them, but governments have made progress in establishing related frameworks.

Frameworks addressing ICT policies often require breadth to address the variety of complex challenges that emerge from the development, management, and integration of technology into all facets of society. A comprehensive ICT policy framework can be developed along seven dimensions: (i) connectivity and access; (ii) usage; (iii) legal and regulatory framework; (iv) production and trade; (v) skills and human resources; (vi) cybersecurity; and (vii) new applications. Table 2 summarizes the key priorities in each dimension of this policy framework.

While the framework in Table 2 includes the most important areas that must be considered by ICT policy-makers, effective formulation and implementation of ICT policy involves several additional issues. First, the vision, strategy, and commitment by government and sector leadership play a crucial role in determining whether and to what extent a country's ICT program produces benefits. This necessitates a new way of thinking, namely the belief that every organization can improve its performance with more effective use of ICT. In particular, through ICT an organization can access more information, gain valuable insights from available data, and solicit input for decision making by more effectively interacting with involved parties.

If a country is below a certain threshold of ICT readiness due to lack of vision, strategy, and commitment on the part of its leaders, it becomes stagnant from the continuation of old development models and may fail to prosper in the information age. Examples of anti-developmental patterns of thinking in ICT policy include reluctance to embrace cloud computing, strict controls on access to data and cross-border data transfer, deterrence of individuals and businesses from adopting the concept of a sharing economy, and lack of coordinated and experimental efforts to promote smart city development.

²¹ Data from World Trade Organization (2015).

²² See: http://pcic.merage.uci.edu/papers/2011/value_ipad_iphone.pdf. Kraemer et al. (2011), however, overlook other benefits reaped by China by participating in the global supply chain of the ICT hardware industry. Among them are the earnings from other sources such as corporate income taxes, land rent, local inputs, and logistic services. At the same time, this participation fosters learning and has paved the way for China's domestic companies to rapidly grow into global players.

Second, governments must learn from international best practices and create test beds or pilot spaces for new policies and programs that enable economies to develop through ICT. Such programs do not necessarily need to be complex, top-down initiatives. In mimicking the agility and adaptability of innovation in the private technology center, government programs must embrace the characteristics of the new knowledge economy: openness, collaboration, and efficiency. One example is Zimbabwe's Hypercube Hub,²³ which is a co-working program that targets innovation start-ups focusing on community impacts; it is also supported financially by the U.S. Embassy in Zimbabwe. These types of programs bring together innovators, investors, and workers and embody the type of economic development that is possible through ICT expansion. Governments should be open to such experimental programs.

Third, governments must exploit the synergy between globalization and the ICT diffusion. Their mutually-reinforcing effects can foster the contribution of ICT to growth. In the increasingly connected economy, the exchange of knowledge and creative ideas across borders and oceans is facilitated by ICT. Communication programs and applications such as Skype and WhatsApp support instantaneous communication. The underlying network infrastructure to support such applications is crucial for enabling the growth of connected industries in the globalized economy. Governments must think several steps ahead in adopting ICT development policies, understanding how a simple but effective ICT foundation can enable future growth in applications and technologies that are impossible to anticipate and plan for today.

Fourth, given that the costs of ICT have over time fallen to more affordable levels, support for ICT use should become a key priority. While reducing the cost of ICT use would be considered a strategic priority, it is more important to increase the benefits that businesses and individuals can reap from investment in ICT applications. The gains from ICT use can be calculated as the difference between its benefits and costs. Bias towards cost reduction may lower the quality and variety of services. Increasing the benefits from ICT use, on the other hand, promotes both ICT use and the quality and variety of ICT services and products. As such, the policy implication is that resources should be allocated more to generating benefits of ICT use than to subsidies for ICT cost reduction. Priority should be given to e-government development, e-business applications, and internet-enabled services.

Finally, governments should consider fundamental reforms in digitizing government services. This type of capability overhaul in government systems, which are often regarded as unwieldy and inefficient, should be based on a robust review of the soundness of all policies and services to determine the feasibility of their delivery through digital (e- or m-) platforms. The cost savings through efficiency generated by government service digitization can create fiscal space to re-invest in additional programs that enhance the benefits of ICT to businesses and individuals. The pioneering role of the government in applying ICT to enhance efficiency and effectiveness, especially in interacting with citizens and businesses, can have strong stimulating effects on the adoption of ICT across sectors of society.

From our five general recommendations we arrive at three conclusions that bear consideration by policy makers. First, the emergence of ICT has permanently reshaped the way businesses is conducted, and therefore governments must make appropriate adjustments in the way policies are designed to manage and support new types of business practices. ICT has also added another dimension to interpersonal connectivity, with greater implications for society, collective action, and public welfare. Governments must stay ahead of these trends to be internationally competitive.

Second, governments are not alone in developing ICT policy. Even while the economy has become increasingly globalized, the knowledge and creativity of generating ICT capabilities have become globalized as well. For example, the development of mobile applications has generated new opportunities for residents of developing countries, particularly in understanding agricultural market conditions, monitoring personal health, and discovering educational opportunities.

Third, while much of the initial technology underlying ICT has originated in developed countries, enterprising residents of developing countries are re-making these technologies in ways that are uniquely beneficial. Governments must recognize and support these trends, understanding that economic growth comes not only from the corporate side, but from small and entrepreneurial ventures that often rely on technology. Fourth, ICT policy is not a one-dimensional effort, but must involve systemic alignment, coordination across government units, and collaboration with business and society. In some countries, this may require an overhaul of policy development processes to integrate non-government interests and concerns. Finally, enhancing the coordination among countries in fostering global digital integration and adopting the latest applications of ICT such as Internet of Things and Industry 4.0 can make the ICT revolution more beneficial. It would be valuable if organizations such as the UN, OECD, World Bank will allocate more resources for this coordination effort, which can range from collecting ICT investment data and diffusing knowledge on ICT applications to envisioning the future digital world and coordinating investments in ICT infrastructure.

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²³ <http://www.hypercubehub.co/en>.

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